

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Vaishnavi.S
Register Number	192521313

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I vaishnavi.s (192521313) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage

Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition

Cloud computing is a technology that provides computing resources such as servers, storage, databases, networking, software, and applications over the Internet on demand. Instead of purchasing and maintaining physical hardware, users can access these resources whenever needed by paying only for what they use. According to the National Institute of Standards and Technology (NIST), cloud computing is a model that enables convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort.

Characteristics of Cloud Computing

1. On-Demand Self-Service

One of the most important characteristics of cloud computing is on-demand self-service. It allows users to access computing resources such as virtual machines, storage, and applications automatically whenever required without the assistance of the cloud service provider. This reduces waiting time and improves efficiency because users can create, modify, or remove resources according to their needs. For example, a company can instantly create a new virtual server using Amazon Web Services (AWS) without contacting technical support.

2. Broad Network Access

Cloud services are available over the Internet and can be accessed through different devices such as laptops, desktops, smartphones, and tablets. Since cloud resources are accessible from anywhere with an Internet connection, users can work remotely and collaborate easily. This feature supports businesses with employees working from different geographical locations and enables continuous access to applications and data.

3. Resource Pooling

Cloud providers maintain a large pool of computing resources that are shared among multiple users. This concept is called resource pooling or multi-tenancy. Physical and virtual resources are dynamically assigned and reassigned based on customer demand. Although multiple users share the same infrastructure, each user's data remains secure and isolated. Resource pooling improves hardware utilization and reduces operational costs.

4. Rapid Elasticity

Cloud computing offers rapid elasticity, which means resources can be increased or decreased quickly according to workload requirements. During periods of high demand, additional servers and storage can be allocated automatically. Similarly, when demand decreases, unnecessary resources can be released. This flexibility allows organizations to avoid purchasing expensive hardware that may remain unused most of the time.

5. High Availability

Cloud providers maintain multiple data centers and backup systems to ensure that services remain available even if one server fails. This minimizes downtime and ensures continuous access to applications and data.

6. Multi-Tenancy

Cloud computing allows multiple customers, known as tenants, to share the same physical infrastructure while keeping their data secure and isolated. This improves resource utilization and reduces operational costs for both providers and users.

Difference from Traditional On-Premise Computing

Traditional on-premise computing requires organizations to purchase, install, and maintain their own hardware, software, and networking equipment. It involves high initial investment and limited scalability. In contrast, cloud computing provides services through the Internet, offers flexible resource allocation, lower maintenance costs, and allows users to pay only for the resources they consume.

Applications of Cloud Computing

Cloud computing is widely used in various sectors. Educational institutions use cloud platforms for online classes and learning management systems. Businesses rely on cloud services for data storage, email, customer relationship management (CRM), and enterprise applications. Healthcare organizations store patient records securely on the cloud, while banks use cloud infrastructure for online banking services. Entertainment platforms such as Netflix and Spotify stream content using cloud computing. Social media platforms like Facebook and Instagram also depend on cloud technologies to serve millions of users worldwide.

Conclusion

Cloud computing has transformed the way organizations use technology by providing flexible, scalable, and cost-effective computing resources. Features such as on-demand self-service, broad network access, resource pooling, and rapid elasticity make cloud computing more efficient and reliable than traditional on-premise computing.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Introduction

Cloud computing did not emerge suddenly; it evolved through several technological advancements over many decades. The development of powerful hardware, networking technologies, virtualization, and Internet-based software gradually led to the creation of modern cloud platforms that provide computing resources as online services.

Hardware Evolution

The evolution of cloud computing began with mainframe computers during the 1950s and 1960s. These were large and expensive systems shared by multiple users through terminals using a technique called time-sharing. This concept allowed efficient utilization of computing resources and introduced the idea of sharing infrastructure among several users.

During the 1980s, personal computers became affordable and widely available. Organizations started purchasing their own computers and running applications locally. Although this provided greater control, it increased the cost of hardware maintenance and software management.

Client-Server Computing

In the 1990s, client-server architecture became popular. In this model, clients requested services from centralized servers through computer networks. Servers handled data storage, application processing, and resource management, while clients focused on user interaction. This architecture improved data sharing, centralized management, and network communication.

Internet Evolution

As the Internet expanded rapidly, businesses started offering services online. Applications such as email, online banking, e-commerce, and web portals became accessible from anywhere in the world. The Internet made global communication faster and encouraged the development of web-based services.

Virtualization

Virtualization technology allowed multiple virtual machines to run on a single physical server. This significantly improved hardware utilization, reduced infrastructure costs, and simplified server management. Virtualization became one of the most important technologies that enabled cloud computing because it allowed cloud providers to efficiently allocate resources to multiple customers.

Internet Software Evolution

Software gradually shifted from desktop applications to web-based applications. Instead of installing software on individual computers, users could access applications directly through web browsers. Services such as Gmail, Google Docs, and Microsoft Office 365 demonstrated the effectiveness of delivering software through the Internet.

Modern Cloud Platforms

The combination of hardware improvements, virtualization, networking, and Internet software led to the development of modern cloud platforms such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). These platforms provide computing power, storage, databases, networking, artificial intelligence, and many other services on demand.

Benefits of the Evolution of Cloud Computing

The evolution of cloud computing has brought several advantages. Organizations can scale resources according to demand without purchasing additional hardware. Infrastructure costs have been significantly reduced because users follow a pay-as-you-go pricing model. Cloud platforms provide high availability, automatic backups, disaster recovery, and improved collaboration.

Conclusion

The evolution of cloud computing represents the combined progress of hardware, networking, virtualization, and Internet technologies. These advancements created the foundation for modern cloud platforms that offer reliable, scalable, and cost-effective computing services to individuals and organizations worldwide.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Introduction

Server virtualization is one of the most important technologies behind cloud computing. It enables multiple virtual servers to run on a single physical server, making better use of hardware resources. Before virtualization, organizations often used one physical server for a single application, which resulted in low hardware utilization and high maintenance costs. Virtualization solved this problem by allowing several independent virtual machines to share the same physical hardware. This technology forms the foundation of modern cloud computing because it enables efficient resource sharing, scalability, and cost reduction.

Role of Server Virtualization in Cloud Computing

Server virtualization is the process of creating multiple virtual machines (VMs) on a single physical server with the help of software called a hypervisor. Each virtual machine behaves like a separate computer with its own operating system, applications, memory, and storage. Although all virtual machines share the same physical hardware, they operate independently without affecting one another. In cloud computing, virtualization plays a vital role by improving hardware utilization.

Virtualization also simplifies backup, migration, and disaster recovery. Virtual machines can easily be copied, moved, or restored from backups, reducing downtime and ensuring business continuity. It also allows cloud providers to perform server maintenance without significantly affecting users.

Overall, server virtualization enables cloud providers to deliver flexible, reliable, and cost-effective services to millions of users around the world.

Difference Between Virtualization and Cloud Computing

Virtualization is a technology that creates virtual versions of physical resources such as servers, storage, and operating systems. Its main objective is to improve resource utilization by allowing multiple virtual machines to run on one physical server. Cloud computing is a service model that delivers computing resources such as servers, storage, networking, software, and databases over the Internet. Cloud computing uses virtualization along with automation, networking, and management tools to provide services to users on demand.

Virtualization mainly focuses on efficient hardware utilization within a data center. Cloud computing focuses on providing scalable and on-demand services to users through the Internet. Virtualization can exist without cloud computing, but cloud computing largely depends on virtualization to provide efficient and flexible services.

Conclusion

Server virtualization is the backbone of cloud computing because it enables efficient utilization of hardware resources, rapid scalability, resource isolation, and simplified management. While virtualization is an enabling technology that creates virtual machines, cloud computing is a service model that delivers these virtualized resources to users over the Internet. Together, they have transformed the way organizations deploy and manage IT infrastructure.

4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.

Introduction

Web services enable communication and data exchange between different software applications over a network. They allow applications developed using different programming languages and platforms to interact with each other. The two most widely used types of web services are SOAP (Simple Object Access Protocol) and REST (Representational State Transfer). Although both are used to build web services, they differ in communication methods, data formats, performance, and suitability for cloud applications.

SOAP-Based Web Services

SOAP is a protocol designed for exchanging structured information between applications. It follows strict standards and uses XML as its message format. Every request and response in SOAP is enclosed within an XML envelope, making communication highly structured and secure. SOAP supports various communication protocols such as HTTP, SMTP, and TCP. One of the major strengths of SOAP is its support for WS-Security, reliable messaging, and transaction management.

REST-Based Web Services

REST is not a protocol but an architectural style for designing web services. It primarily uses standard HTTP methods such as GET, POST, PUT, and DELETE to perform operations on resources. Unlike SOAP, REST is lightweight and supports multiple data formats, including JSON, XML, HTML, and plain text. Among these, JSON is the most commonly used because it is compact and easy to process.

Comparison Between SOAP and REST

SOAP follows a protocol-based communication style with strict rules, while REST follows an architectural style based on standard HTTP methods. SOAP exchanges data only in XML format, whereas REST supports multiple formats such as JSON and XML. SOAP provides advanced security features like WS-Security and reliable messaging, making it suitable for banking systems, financial transactions, and enterprise applications.

Suitability for Cloud Applications

Cloud applications require services that are fast, scalable, lightweight, and capable of handling a large number of requests efficiently. REST satisfies these requirements by using simple HTTP communication and compact JSON data formats. It reduces network traffic and improves application performance, making it ideal for cloud environments. SOAP is still preferred in situations where security, reliability, and transactional support are more important than speed, such as banking systems, healthcare applications, and enterprise business services.

Conclusion

Both SOAP and REST are important web service technologies with different strengths. SOAP provides high security, reliability, and standardized communication, making it suitable for enterprise applications. REST offers simplicity, speed, flexibility, and better scalability, making it the preferred choice for modern cloud computing and web-based applications. Therefore, REST is widely adopted in today's cloud platforms, while SOAP remains important for secure business-critical systems.

